Agile Fat
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<u>App.java</u>

```
package com.example;

import java.util.Random;

public class App {

    private String generatedOTP;

    // Generate OTP
    public String generateOTP() {
        Random rand = new Random();
        int otp = 1000 + rand.nextInt(9000); // 4-digit OTP
        generatedOTP = String.valueOf(otp);
        return generatedOTP;
    }

    // Validate OTP
    public boolean validateOTP(String inputOTP) {
        return generatedOTP != null && generatedOTP.equals(inputOTP);
    }

    // Reset password
    public boolean resetPassword(String inputOTP, String newPassword) {

        // Password rule: at least 6 characters
        if (newPassword == null || newPassword.length() < 6) {
            return false;
        }

        // OTP must be valid
        if (!validateOTP(inputOTP)) {
            return false;
        }

        return true; // reset success
    }

    public static void main(String[] args) {
        App app = new App();

        String otp = app.generateOTP();
        System.out.println("Generated OTP: " + otp);

        boolean result = app.resetPassword(otp, "newpass123");

        if (result) {
```

```
            System.out.println("Password Reset Successful");
        } else {
            System.out.println("Password Reset Failed");
        }
    }
}
```

AppTest.java

```
package com.example;

import org.junit.Test;
import static org.junit.Assert.*;

public class AppTest {

    @Test
    public void testSuccessfulReset() {
        App app = new App();
        String otp = app.generateOTP();

        assertTrue(app.resetPassword(otp, "secure123"));
    }

    @Test
    public void testInvalidOTP() {
        App app = new App();
        app.generateOTP();

        assertFalse(app.resetPassword("0000", "secure123"));
    }

    @Test
    public void testWeakPassword() {
        App app = new App();
        String otp = app.generateOTP();

        assertFalse(app.resetPassword(otp, "123"));
    }
}
```

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0">
    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>password-reset-app</artifactId>
    <version>1.0</version>

    <dependencies>
```

```
        <dependency>
            <groupId>junit</groupId>
            <artifactId>junit</artifactId>
            <version>4.13.2</version>
            <scope>test</scope>
        </dependency>
    </dependencies>

    <build>
        <plugins>

            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.10.1</version>
                <configuration>
                    <source>17</source>
                    <target>17</target>
                </configuration>
            </plugin>

            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-jar-plugin</artifactId>
                <version>3.2.2</version>
                <configuration>
                    <archive>
                        <manifest>
                            <mainClass>com.example.App</mainClass>
                        </manifest>
                    </archive>
                </configuration>
            </plugin>

        </plugins>
    </build>
</project>
```

Pipeline

```
pipeline {
    agent any

    tools {
        maven 'M3'
    }

    stages {

        stage('Checkout Git') {
```

```
        steps {
            git branch: 'main',
                url: 'https://github.com/aishwaryasuresh020-source/agile.git'
        }
    }

    stage('Build and Test') {
        steps {
            sh 'mvn clean test'
        }
    }

  }
}
```

Dockerfile

```
FROM eclipse-temurin:17-jdk

WORKDIR /app

COPY target/password-reset-app-1.0.jar app.jar

CMD ["java", "-jar", "app.jar"]
```

deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: password-reset-app

spec:
  replicas: 2

  selector:
    matchLabels:
      app: password-reset

  template:
    metadata:
      labels:
        app: password-reset

    spec:
      containers:
      - name: password-reset-container
```

```
      image: yourdockerhubusername/password-reset-app:latest
---
apiVersion: v1
kind: Service
metadata:
  name: password-reset-service

spec:
  type: NodePort

  selector:
    app: password-reset

  ports:
    - port: 80
      targetPort: 8080
      nodePort: 30011
```

Screenshots:

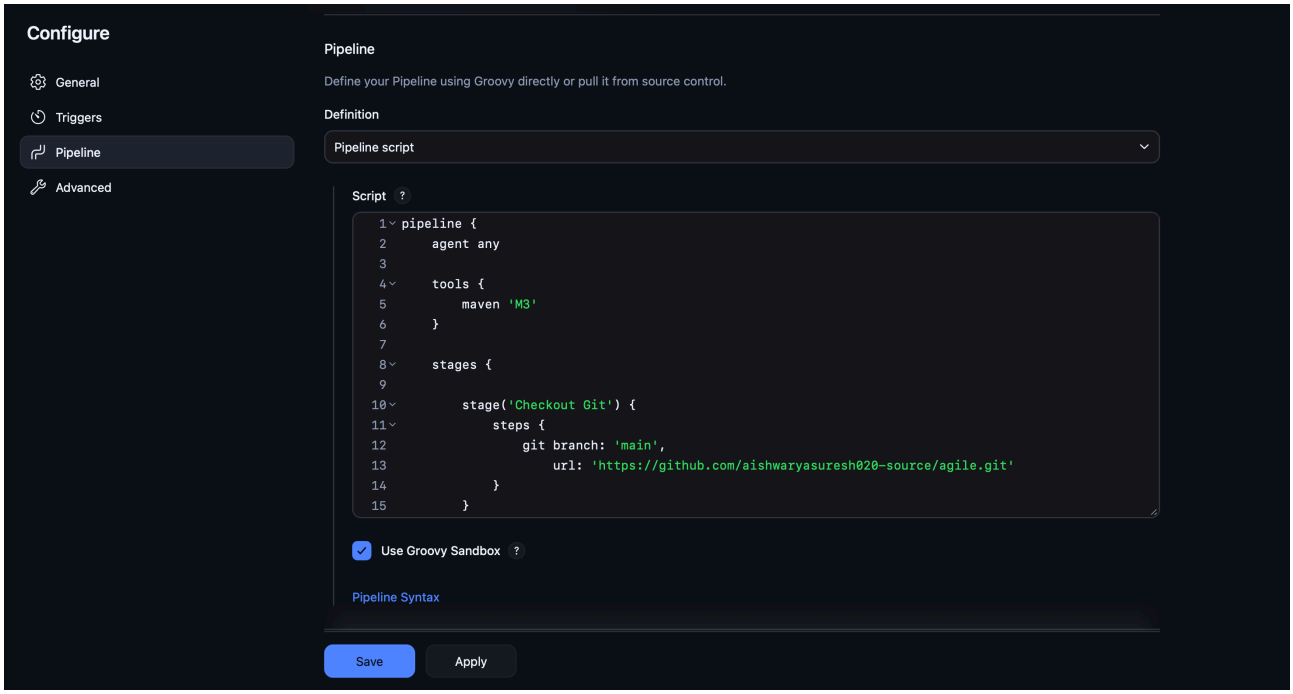
Git:

```
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile % git init
Initialized empty Git repository in /Users/aishwaryasuresh/Downloads/Agile/.git/
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile % git clone https://github.com/aishwaryasuresh020-source/agile.git
Cloning into 'agile'...
remote: Enumerating objects: 22, done.
remote: Counting objects: 100% (22/22), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 22 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (22/22), 4.96 KiB | 2.48 MiB/s, done.
Resolving deltas: 100% (2/2), done.
```

JUnit Test:

```
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile % cd Agile
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile % mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:password-reset-app >-------------------
[INFO] Building password-reset-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ password-reset-app ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ password-reset-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /Users/aishwaryasuresh/Downloads/Agile/agile/src/main/resources
[INFO]
[INFO] --- compiler:3.10.1:compile (default-compile) @ password-reset-app ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /Users/aishwaryasuresh/Downloads/Agile/agile/target/classes
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ password-reset-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /Users/aishwaryasuresh/Downloads/Agile/agile/src/test/resources
[INFO]
[INFO] --- compiler:3.10.1:testCompile (default-testCompile) @ password-reset-app ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /Users/aishwaryasuresh/Downloads/Agile/agile/target/test-classes
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ password-reset-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.019 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.702 s
[INFO] Finished at: 2026-04-17T16:37:08+05:30
[INFO] ------------------------------------------------------------------------
```

Building Pipeline in Jenkins:



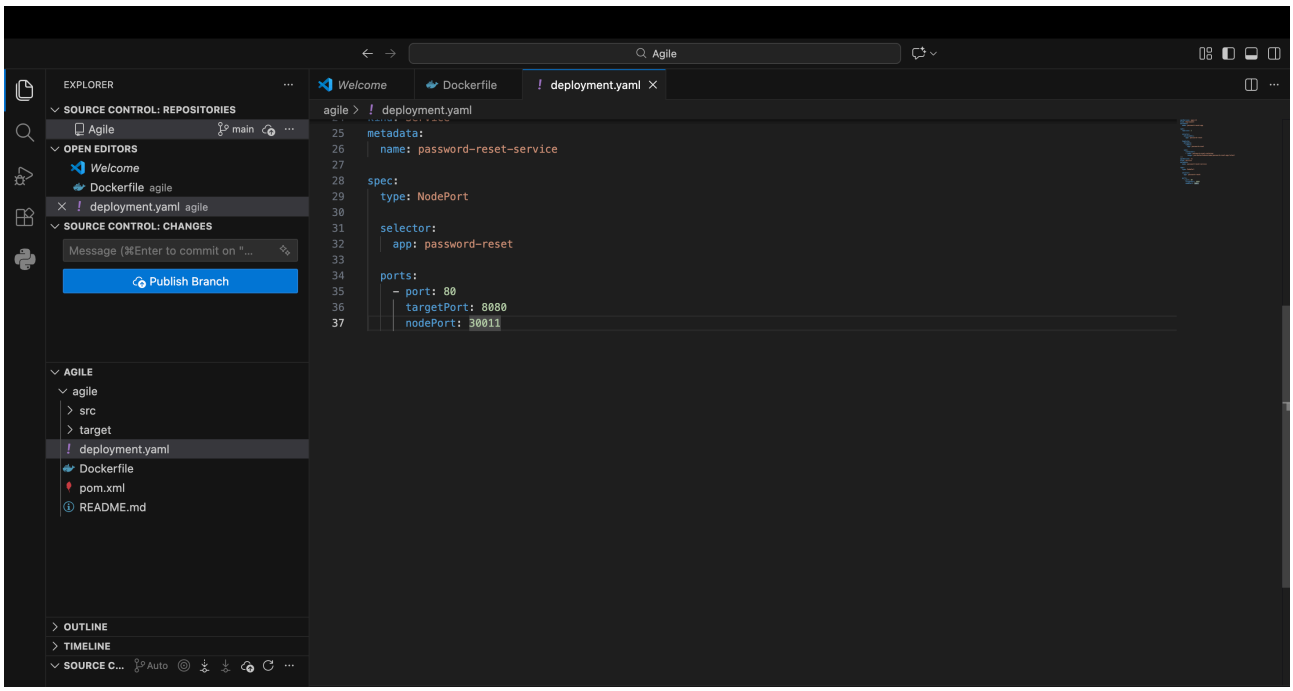
Jenkins Pipeline Output:

```
Started by user Aishwarya Suresh
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /Users/aishwaryasuresh/.jenkins/workspace/agile_fat
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Tool Install)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Checkout Git)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/aishwaryasuresh020-source/agile.git
 > git init /Users/aishwaryasuresh/.jenkins/workspace/agile_fat # timeout=10
Fetching upstream changes from https://github.com/aishwaryasuresh020-source/agile.git
 > git --version # timeout=10
 > git --version # 'git version 2.39.5 (Apple Git-154)'
 > git fetch --tags --force --progress -- https://github.com/aishwaryasuresh020-source/agile.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
 > git config remote.origin.url https://github.com/aishwaryasuresh020-source/agile.git # timeout=10
 > git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
```

```
 > git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 7b654c83563d3d54af8b6da024a1aa6f9c76a6d9 (refs/remotes/origin/main)
 > git config core.sparsecheckout # timeout=10
 > git checkout -f 7b654c83563d3d54af8b6da024a1aa6f9c76a6d9 # timeout=10
 > git branch -a -v --no-abbrev # timeout=10
 > git checkout -b main 7b654c83563d3d54af8b6da024a1aa6f9c76a6d9 # timeout=10
Commit message: "Create AppTest.java"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build and Test)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] sh
+ mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:password-reset-app >-------------------
[INFO] Building password-reset-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ password-reset-app ---
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ password-reset-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /Users/aishwaryasuresh/.jenkins/workspace/agile_fat/src/main/resources
[INFO]
```

```
[INFO] Compiling 1 source file to /Users/aishwaryasuresh/.jenkins/workspace/agile_fat/target/test-classes
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ password-reset-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.018 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.690 s
[INFO] Finished at: 2026-04-17T16:45:46+05:30
[INFO] ------------------------------------------------------------------------
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Building Dockerfile and deployment.yaml



Mvn clean package:

```
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile % mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:password-reset-app >-------------------
[INFO] Building password-reset-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ password-reset-app ---
[INFO] Deleting /Users/aishwaryasuresh/Downloads/Agile/agile/target
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ password-reset-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /Users/aishwaryasuresh/Downloads/Agile/agile/src/main/resources
[INFO]
[INFO] --- compiler:3.10.1:compile (default-compile) @ password-reset-app ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /Users/aishwaryasuresh/Downloads/Agile/agile/target/classes
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ password-reset-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /Users/aishwaryasuresh/Downloads/Agile/agile/src/test/resources
[INFO]
[INFO] --- compiler:3.10.1:testCompile (default-testCompile) @ password-reset-app ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /Users/aishwaryasuresh/Downloads/Agile/agile/target/test-classes
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ password-reset-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.018 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.2.2:jar (default-jar) @ password-reset-app ---
[INFO] Building jar: /Users/aishwaryasuresh/Downloads/Agile/agile/target/password-reset-app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.850 s
[INFO] Finished at: 2026-04-17T16:37:30+05:30
[INFO] ------------------------------------------------------------------------
```

Implementing Dockerfile:

```
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile % docker build -t aishwaryasuresj/password-reset-app .
[+] Building 0.1s (8/8) FINISHED                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                0.0s
 => => transferring dockerfile: 159B                                                                                0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk                                           0.0s
 => [internal] load .dockerignore                                                                                   0.0s
 => => transferring context: 2B                                                                                     0.0s
 => [1/3] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:2ff0b91aaa4cb5dc35b4760c37fe064390a486dcd90b36  0.0s
 => => resolve docker.io/library/eclipse-temurin:17-jdk@sha256:2ff0b91aaa4cb5dc35b4760c37fe064390a486dcd90b36  0.0s
 => [internal] load build context                                                                                   0.0s
 => => transferring context: 3.04kB                                                                                 0.0s
 => CACHED [2/3] WORKDIR /app                                                                                       0.0s
 => [3/3] COPY target/password-reset-app-1.0.jar app.jar                                                            0.0s
 => exporting to image                                                                                              0.0s
 => => exporting layers                                                                                             0.0s
 => => exporting manifest sha256:91532713f79b41e4ea49557d87ae28dc442186442ccb8c89ccdc20320c704f4c                   0.0s
 => => exporting config sha256:508e71bc246b201318f1322935a86ed0c9edc923fc1e398eff5f73dd953da675                     0.0s
 => => exporting attestation manifest sha256:7c96d6aa1565b46e78666bd350312522d80ec4ed062d9c9e2a5da26bdfb07bc6  0.0s
 => => exporting manifest list sha256:23d08def2974fc0fdde534bd0268b03ac38a8bf35c41b285c5f900a9a53b3aaa         0.0s
 => => naming to docker.io/aishwaryasuresj/password-reset-app:latest                                                0.0s
 => => unpacking to docker.io/aishwaryasuresj/password-reset-app:latest                                             0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/ku2m5esqnbxge1sj65rwu2rl7
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile % docker login
Authenticating with existing credentials... [Username: aishwaryasuresj]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile % docker push aishwaryasuresj/password-reset-app
Using default tag: latest
The push refers to repository [docker.io/aishwaryasuresj/password-reset-app]
af0cae6e26c9: Mounted from aishwaryasuresj/shopping-cart-app
76fd055477b6: Mounted from aishwaryasuresj/shopping-cart-app
ec83fb152f23: Mounted from aishwaryasuresj/shopping-cart-app
59b4fba6561b: Mounted from aishwaryasuresj/shopping-cart-app
4e985483d507: Mounted from aishwaryasuresj/shopping-cart-app
62c8606cc493: Mounted from aishwaryasuresj/shopping-cart-app
6862eab6b1fc: Pushed
a2803411f161: Pushed
latest: digest: sha256:23d08def2974fc0fdde534bd0268b03ac38a8bf35c41b285c5f900a9a53b3aaa size: 856
```

Kubernetes deployment and service YAML files:

```
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile % kubectl apply -f deployment.yaml
deployment.apps/password-reset-app created
service/password-reset-service created
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile % kubectl get pods
NAME                                  READY   STATUS              RESTARTS         AGE
order-app-deployment-74bdc75d97-zf97d   0/1     ErrImageNeverPull   0                4d18h
password-reset-app-d584974f7-hwr8m      0/1     ErrImagePull        0                6s
password-reset-app-d584974f7-tzflq      0/1     ContainerCreating   0                6s
payment-app-6b9978f854-5xqks            0/1     ImagePullBackOff    0                4d6h
payment-app-6b9978f854-r456d            0/1     ImagePullBackOff    0                4d6h
refund-app-8fb9cd9db-f6b7n              0/1     ImagePullBackOff    0                4d12h
refund-app-8fb9cd9db-qj25f              0/1     ImagePullBackOff    0                4d12h
resume-deployment-84c4cc5dfc-kqzcz      1/1     Running             12 (3m12s ago)   11d
resume-deployment-84c4cc5dfc-xgz9r      1/1     Running             12 (3m12s ago)   11d
shopping-cart-app-75b4d8c96b-5lcrn      0/1     Error               47 (92s ago)     4d14h
shopping-cart-app-75b4d8c96b-vv4t7      0/1     CrashLoopBackOff    45 (94s ago)     4d14h
aishwaryasuresh@Aishwaryas-MacBook-Air-2 Agile % kubectl get svc
NAME                     TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
kubernetes               ClusterIP   10.96.0.1       <none>        443/TCP        12d
password-reset-service   NodePort    10.96.115.182   <none>        80:30011/TCP   13s
payment-service          NodePort    10.96.201.188   <none>        80:30099/TCP   4d6h
refund-service           NodePort    10.96.59.111    <none>        80:30003/TCP   4d12h
resume-service           NodePort    10.96.58.224    <none>        80:30007/TCP   11d
shopping-cart-app        NodePort    10.96.171.189   <none>        80:30009/TCP   4d14h
```

Docker Image:

Images Give feedback

Local My Hub

536.48 MB / 3.61 GB in use 15 images

Last refresh: 20 minutes ago

Search

Name	Tag	Image ID	Created	Size	Actions
aishwaryasuresh/order-app	latest	9e620d160d54	5 days ago	667.34 MB	
order-app	latest	9e620d160d54	5 days ago	667.34 MB	
<none>	<none>	104ed9be9700	5 days ago	667.33 MB	
<none>	<none>	ecfcaebc702a	5 days ago	667.33 MB	
eclipse-temurin	17-jdk	2ff0b91aaa4c	10 days ago	671 MB	
contact-service	latest	c450182c9bde	11 days ago	667.33 MB	
resume-app	latest	f08424da7972	12 days ago	92.17 MB	
yourdockerhub/resume-app	latest	7e0e4f0623ac	12 days ago	92.08 MB	
kindest/node	v1.34.3	08497ee19eac	4 months ago	1.32 GB	
docker/desktop-cloud-provider-l	v0.5.0	4ad59ce20658	4 months ago	574.06 MB	
envoyproxy/envoy	v1.36.4	08acc53db6f4	4 months ago	233.28 MB	
docker/desktop-containerd-regi:	v0.0.3	179a9ea3ae47	4 months ago	48.81 MB	
aishwaryasuresj/password-rese	latest	23d08def2974	19 minutes ag	667.34 MB	

Showing 16 items

Kubernete:

Kubernetes Give feedback

Namespace
default

Cluster

Stop Edit cluster

Cluster	Cluster type	Nodes	Kubernetes version
Active	kind	1	v1.34.3

Deployments

Name	Status	Pods
payment-app		0/2
refund-app		0/2
resume-deployment	Available	2/2
shopping-cart-app		0/2
password-reset-app		0/2

Pods

Name	Status
resume-deployment-84c4cc5dfc-xg29r	Running
shopping-cart-app-75b4d8c96b-5lcrn	Running
shopping-cart-app-75b4d8c96b-vv4t7	Running
password-reset-app-d584974f7-tzflq	Pending
password-reset-app-d584974f7-hwr8m	Pending

Nodes

Name	Status

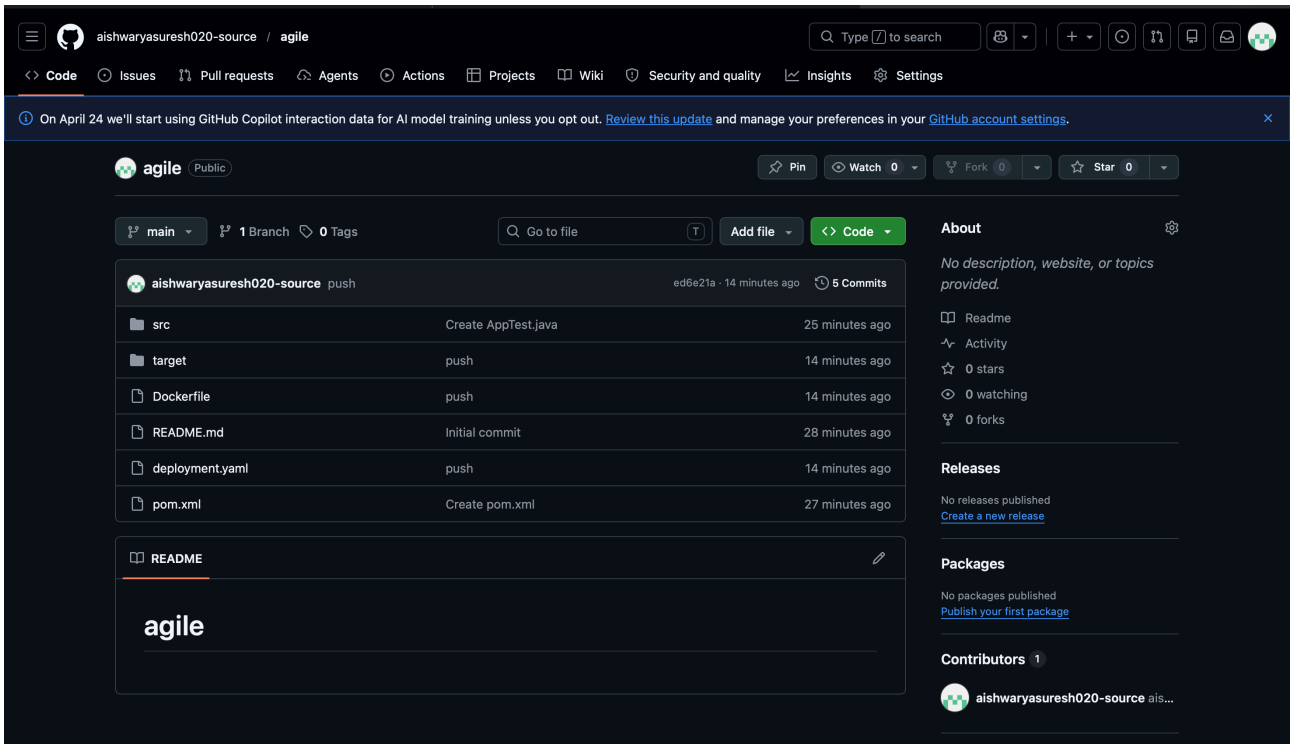
Services

Name	Cluster IP	Ports

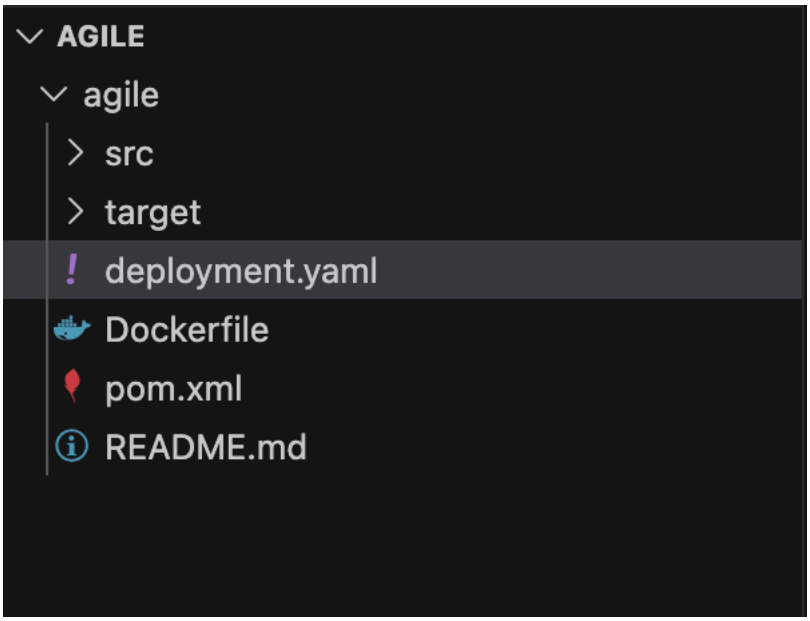
Kubernetes running RAM 2.27 GB CPU 1.21% Disk: 6.06 GB used (limit 223.63 GB)

Terminal Update available

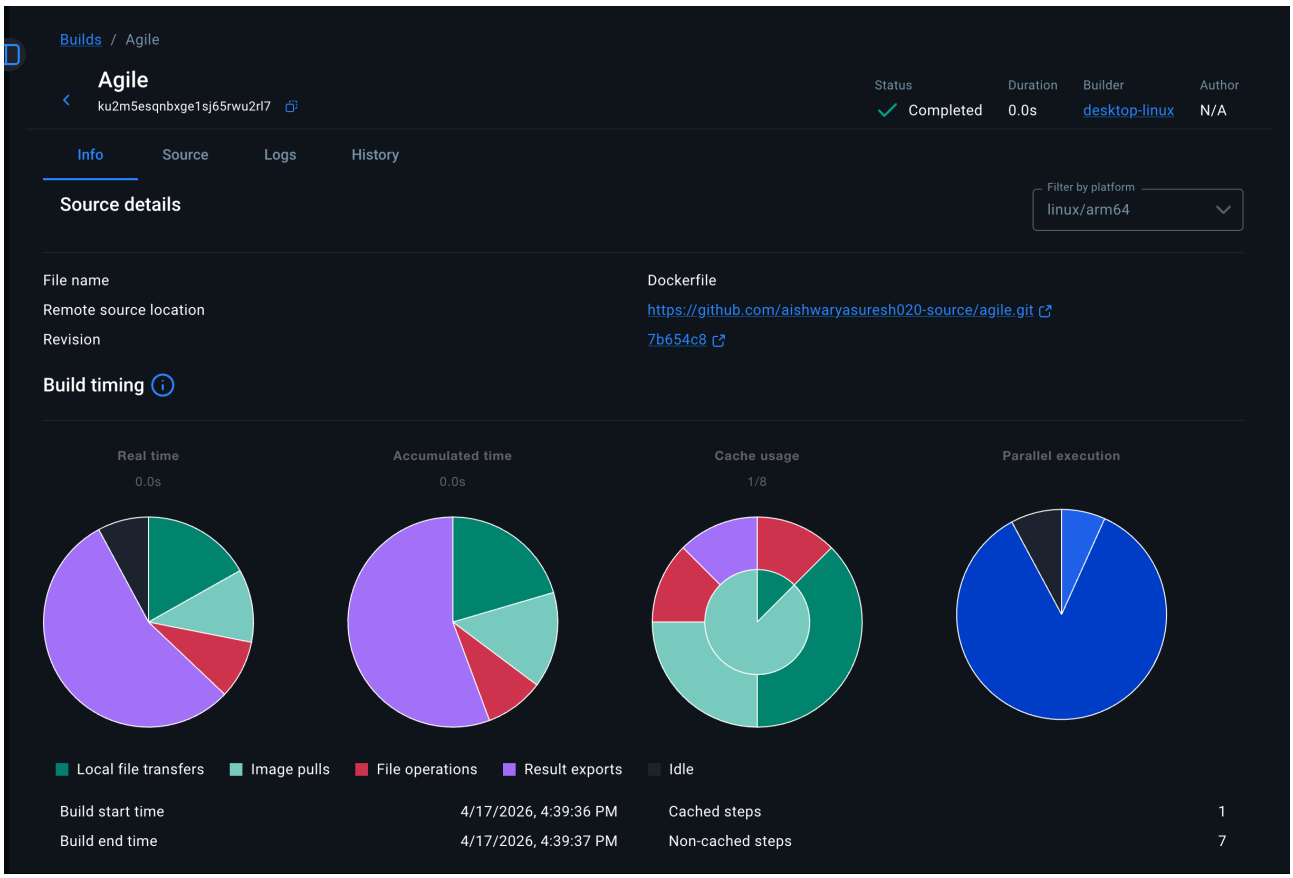
Git structure:

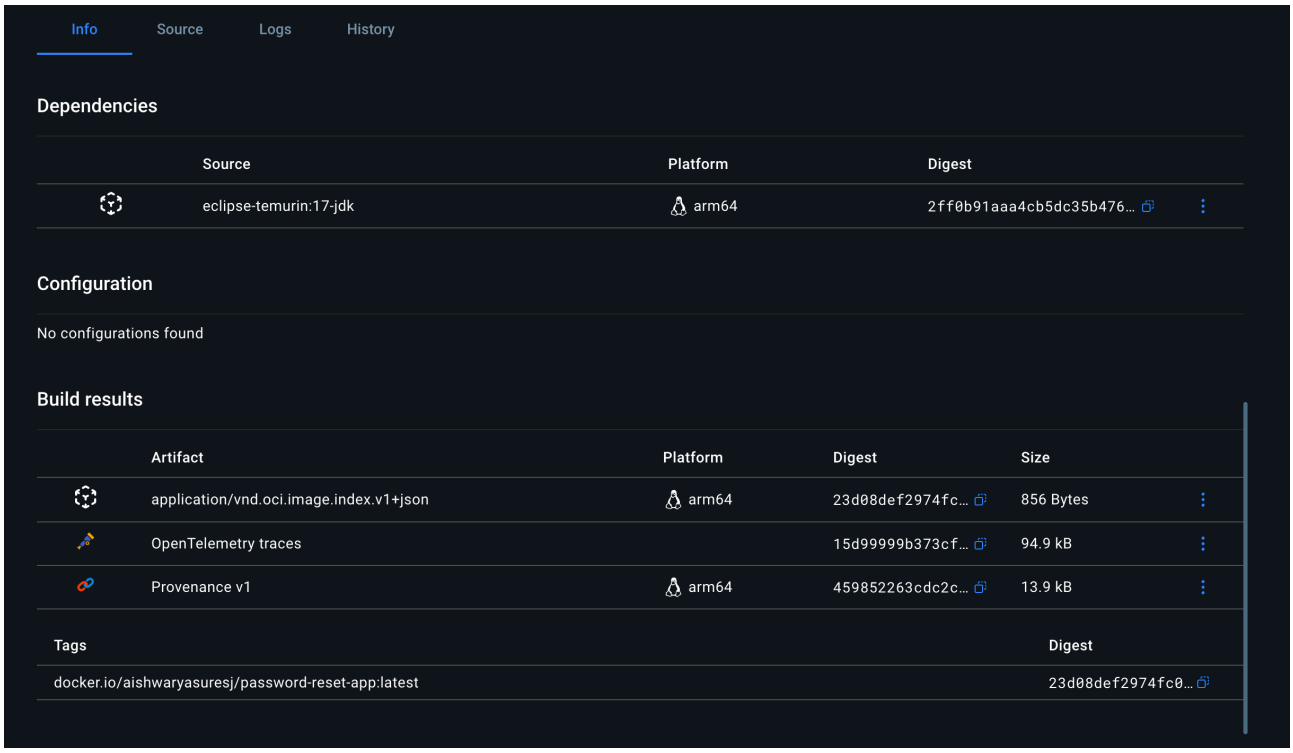


Folder structure:



Build:


Builds / Agile
Agile
ku2m5esqnbxge1sj65rwu2rl7
Status
Completed
Duration
0.0s
Builder
desktop-linux
Author
N/A
Info
Source
Logs
History
Source details
Filter by platform
linux/arm64
File name
Dockerfile
Remote source location
https://github.com/aishwaryasuresh020-source/agile.git
Revision
7b654c8
Build timing
Real time
0.0s
Accumulated time
0.0s
Cache usage
1/8
Parallel execution
Local file transfers
Image pulls
File operations
Result exports
Idle
Build start time
4/17/2026, 4:39:36 PM
Build end time
4/17/2026, 4:39:37 PM
Cached steps
1
Non-cached steps
7


Info
Source
Logs
History
Dependencies
Source
Platform
Digest
eclipse-temurin:17-jdk
arm64
2ff0b91aaa4cb5dc35b476...
Configuration
No configurations found
Build results
Artifact
Platform
Digest
Size
application/vnd.oci.image.index.v1+json
arm64
23d08def2974fc...
856 Bytes
OpenTelemetry traces
15d99999b373cf...
94.9 kB
Provenance v1
arm64
459852263cdc2c...
13.9 kB
Tags
Digest
docker.io/aishwaryasuresj/password-reset-app:latest
23d08def2974fc0...